

Examiner
only

5. (a) Radon gas decays by emitting α -particles. It has a half-life of 3.8 days. Calculate the **percentage reduction** in the activity of a sample of radon after 12 days. [4]

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- (b) A student makes the following measurements for a radioactive source using the indicated absorber between the source and detector.

Absorber	Counts per minute
none	1 004
sheet of paper	597
2 mm of aluminium	23
15 cm of lead	27

Explain these observations. [4]

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